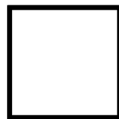
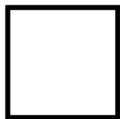
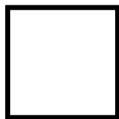
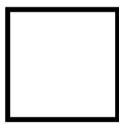
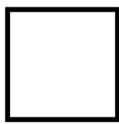


Homework 7

Instructions

- (1) Download the printable version of this assignment [here](#). Print it out and record your answers directly in the space provided.
- (2) Write your **name** and **student number** on the top of the assignment.
- (3) For problems requiring additional written work, use the allotted work space and/or margins of the page.
- (4) Once you're done with a problem, reflect on how you well-equipped you felt answering that particular problem using the confidence level assessment.





(5) Turn in your assignment in class on the due date.

Questions

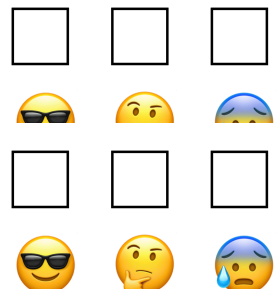
Question 1

A researcher collected the following data on years of education (X) and number of children (Y) for a sample of married adults:

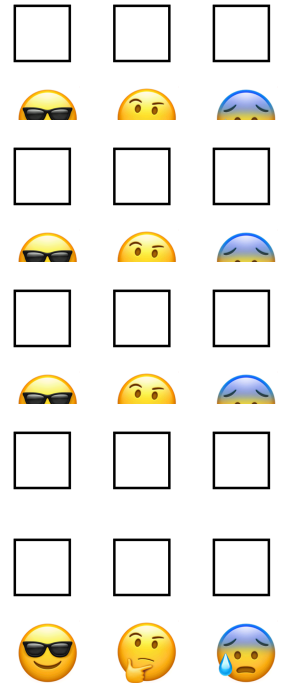
X	Y
12	2
14	1
17	0
10	3
8	5
9	3
12	4
14	2
18	0
16	2

a. Draw a scatter plot of the data.

b. Calculate and interpret the correlation between education and number of children.



- c. Is the association between education and number of children statistically significant? Explain the basis of your answer and the implications for the association in the population.
- d. Calculate and interpret the meaning of the regression slope.
- e. Calculate and interpret the meaning of the Y-intercept for the regression line.
- f. Predict the number of children for an adult with 11 years of education.
- g. Calculate and interpret the coefficient of determination.



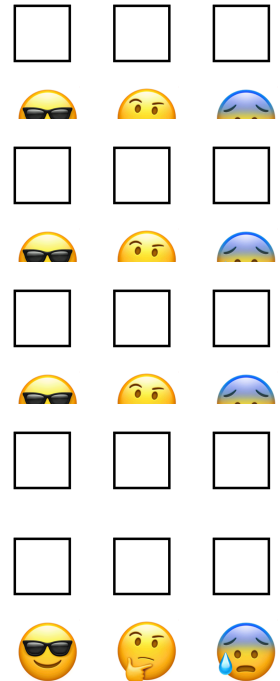
Question 2

The same researcher collected at random one child from each of the eight families in which there was at least one child. She calculated the following statistics to study the effects of number of siblings (X) on happiness of the child (Y), measured on an interval scale in which low scores indicate a low level of happiness and high scores indicate a high level of happiness.

Happiness points (Y)	Number of siblings (X)	
$\bar{y} = 5.625$	$\bar{x} = 1.750$	$r = 0.588$
$s_y = 2.134$	$s_x = 1.282$	$n = 8$

- a. Is the association between number of siblings and happiness statistically significant? Explain the basis of your answer and the implications for the association in the population.

- b. Calculate and interpret the meaning of the regression slope.
- c. Calculate and interpret the meaning of the Y-intercept for the regression line.
- d. Predict the level of happiness for an only child and for a kid with two siblings.
- e. Calculate and interpret the coefficient of determination.



Question 3

A personnel specialist with a large accounting firm is interested in determining the effect of seniority (the number of years with the company) on hourly wages for administrative staff. She selects a random sample of 10 secretaries and collects the following data on their years with the company (X) and hourly wages (Y):

Hourly Wage (Y)	Years with company (X)	
$\bar{y} = \$13.90$	$\bar{x} = 2.700$	$r = 0.631$
$s_y = \$1.37$	$s_x = 1.889$	$n = 10$

- a. Is the association statistically significant? Explain your answer.
- b. Calculate and interpret the meaning of the regression slope.

